MDMC
Master in Data Management and Curation

AREA SCIENCE PARK

SISSA

Python II: Packaging Python code

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Goals

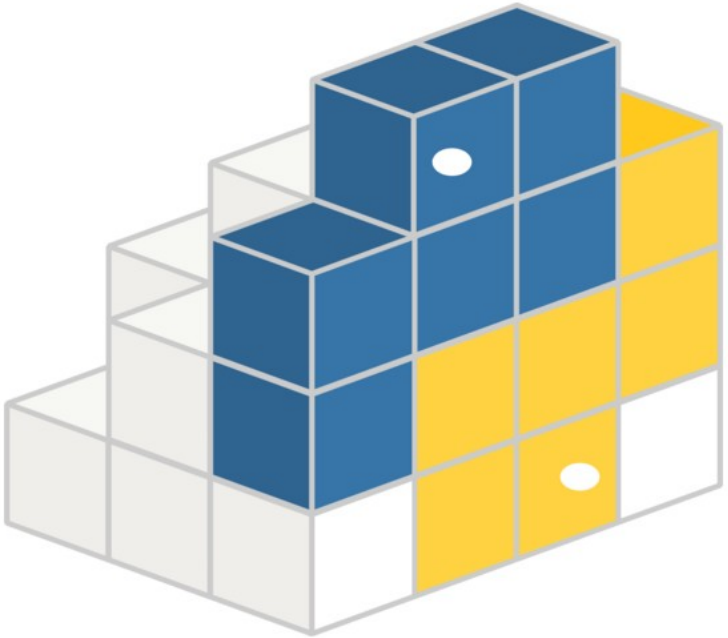


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Objectives of the lecture:

* Learn how to **package** python code
* Import that code to your environment with **`pip`**

Packages

Python packages are **folders** containing one or more **modules** that can be installed with **`pip`** (or `conda`, ...) and imported in other codes.

Example: importing a package

```
>>> import numpy as np
>>> from matplotlib import pyplot as plt
>>> import numpy.fft
```

Creating a package

In order to create a python package, modules have to be organized in a **certain directory structure**

Example: the directory structure

```
improc/
├── LICENSE
├── pyproject.toml
├── README.md
└── src
    └── improc
        ├── images.py
        ├── __init__.py
        └── tools.py
```

The LICENSE

The license file specifies **how others can use your code**. If none is provided, nobody could technically do anything with it*!

You can choose an open source license or any other formula you like.

I usually go with the MIT one.

*except for OpenAI, Anthropic, ...

The README.md

The README.md file provide **information on the package**.

Usually, it contains a short presentation on the functionality and possibly a short tutorial on how to import it.

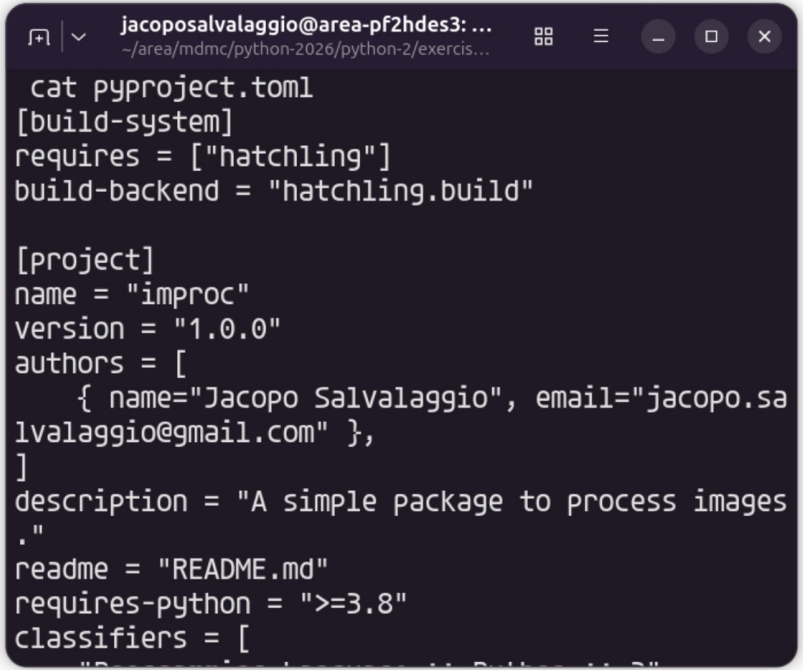


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The pyproject.toml

Contains technical specifications on how to create the package with:

* names
* versions
* dependencies
* boilerplate
* ...



`__init__.py`

As for classes, it **defines what is run when the module is imported**.

It usually contains some documentation and import statements: all the imports done in `__init__.py` will be carried over when we run `import <module name>`

Example: example `__init__.py`

```
"""Package for doing stuff"""
from .tools import some_function
from .classes import MyClass
```

`pip` installing

The main ways to publish your python package are via **PyPi** or with a normal **repo**.

In the former case, the end user should just run `pip install <package name>`

In the second, the user should clone the repo and then run `pip install [-e] ./<repo name>`

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Thank you for your attention :)

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